

Vault — 15 Rounds (Corrected)

Correction made: Round 6, Section 4. The original answer 6 and vault code 3496 were corrected to 8 and 3498.

Round 1

Section 1 — Linear Equations

Solve for x: $4(x - 2) - 3(x - 1) = 2$

Answer: 7

Section 2 — Percentages

A number decreases by 10% to give 90. Find 1% of the original number.

Solution: Original = 100, 1% = 1

Answer: 1

Section 3 — Grid Direction & Distance

Starting at her house facing North, Priya walks a sequence of legs, turning left/right relative to her current facing each time: Walks 9 km (facing North). Turns left, walks 2 km. Turns left, walks 4 km. Turns right, walks 9 km. Turns left, walks 5 km. Turns left, walks 4 km. What is the shortest distance (km, rounded to nearest whole number) between her final position and the start? (If more than 9, give the last digit.)

Solution: Tracking heading and (x,y): final position = (-7,0). Distance = $\sqrt{(-7)^2 + 0^2} = 7.00 \rightarrow$ rounds to 7.

Answer: 7

Section 4 — Ratio & Mixture

A fruit punch contains orange juice and pineapple juice in the ratio 5:3. After 4 L of pineapple juice is added, the ratio becomes 5:4. Find the original quantity of orange juice.

Solution: Let original quantities be 5x and 3x. Then $5x/(3x+4)=5/4$, so $20x=15x+20$, $x=4$. Orange juice=20 L.

Answer: 0

Vault Code (Round 1): 7170

Round 2

Section 1 — Linear Equations

Solve for x: $(x + 2)/2 + (x + 2)/4 = 6$

Answer: 6

Section 2 — Percentages

A sum of ₹200 grows to ₹236 in a year at simple interest. Find 10% of the interest earned (rounded).

Solution: Interest = 36, 10% = $3.6 \approx 4$

Answer: 4

Section 3 — Grid Direction & Distance

Starting at her house facing North, Priya walks a sequence of legs, turning left/right relative to her current facing each time: Walks 3 km (facing North). Turns right, walks 3 km. Turns left, walks 5 km. Turns left, walks 4 km.

Turns left, walks 3 km. Turns left, walks 7 km. Turns right, walks 2 km. What is the shortest distance (km, rounded to nearest whole number) between her final position and the start? (If more than 9, give the last digit.)

Solution: Tracking heading and (x,y): final position = (6,3). Distance = $\sqrt{6^2 + 3^2} = 6.71 \rightarrow$ rounds to 7.

Answer: 7

Section 4 — Ratio & Mixture

A school has boys and girls in the ratio 7:5. When 12 girls join, the ratio becomes 7:6. How many students were there originally?

Solution: Let boys = $7x$, girls = $5x$. $7x/(5x+12) = 7/6$, so $42x = 35x + 84$, $x = 12$. Total = $12x = 144$.

Answer: 4

Vault Code (Round 2): 6474

Round 3

Section 1 — Linear Equations

Solve for x: $2(3x - 1) - 3(x - 2) = 10$

Answer: 2

Section 2 — Percentages

A number is 20% more than 15. Find 20% of the larger number (rounded).

Solution: Larger number = 18, 20% = $3.6 \approx 4$

Answer: 4

Section 3 — Grid Direction & Distance

Starting at her house facing North, Priya walks a sequence of legs, turning left/right relative to her current facing each time: Walks 2 km (facing North). Turns right, walks 2 km. Turns right, walks 7 km. Turns left, walks 1 km. Turns left, walks 4 km. What is the shortest distance (km, rounded to nearest whole number) between her final position and the start? (If more than 9, give the last digit.)

Solution: Tracking heading and (x,y): final position = (3,-1). Distance = $\sqrt{3^2 + 1^2} = 3.16 \rightarrow$ rounds to 3.

Answer: 3

Section 4 — Ratio & Mixture

A bag contains red and blue beads in the ratio 4:7. After 10 red beads are added, the ratio becomes 2:3. How many beads were originally in the bag?

Solution: Let red=4x, blue=7x. $(4x+10)/(7x)=2/3$, so $12x+30=14x$, $x=15$. Total=11x=165.

Answer: 5

Vault Code (Round 3): 2435

Round 4

Section 1 — Linear Equations

Solve for x: $(2x + 1)/3 + (x - 2)/2 = 4$

Answer: 4

Section 2 — Percentages

60% of a class of 45 students passed. Find the number of failed students divided by 3.

Solution: Failed = 18, $18 \div 3 = 6$

Answer: 6

Section 3 — Grid Direction & Distance

Starting at her house facing North, Priya walks a sequence of legs, turning left/right relative to her current facing each time: Walks 7 km (facing North). Turns right, walks 4 km. Turns left, walks 5 km. Turns right, walks 1 km. Turns left, walks 3 km. What is the shortest distance (km, rounded to nearest whole number) between her final position and the start? (If more than 9, give the last digit.)

Solution: Tracking heading and (x,y): final position = (5,15). Distance = $\sqrt{5^2 + 15^2} = 15.81 \rightarrow$ rounds to 16

$\rightarrow$ last digit 6.

Answer: 6

Section 4 — Ratio & Mixture

A paint mixture has red and white paint in the ratio 3:5. If 6 L of red paint is added, the ratio becomes 1:1. Find the

original amount of paint.

Solution: Let red=3x, white=5x. $(3x+6)=5x$, so $x=3$. Original total=8x=24 L.

Answer: 4

Vault Code (Round 4): 4664

Round 5

Section 1 — Linear Equations

Solve for x: $3(2x - 1) = 2(2x + 4) - 5$

Answer: 3

Section 2 — Percentages

25% of a number equals 20. Find 10% of that number.

Solution: Number = 80, 10% = 8

Answer: 8

Section 3 — Grid Direction & Distance

Starting at her house facing North, Priya walks a sequence of legs, turning left/right relative to her current facing each time: Walks 2 km (facing North). Turns left, walks 4 km. Turns left, walks 6 km. Turns right, walks 8 km.

Turns left, walks 8 km. What is the shortest distance (km, rounded to nearest whole number) between her final position and the start? (If more than 9, give the last digit.)

Solution: Tracking heading and (x,y): final position = (-12,-12). Distance = $\sqrt{(-12)^2 + (-12)^2} = 16.97 \rightarrow$ rounds to 17 $\rightarrow$ last digit 7.

Answer: 7

Section 4 — Ratio & Mixture

The ratio of A's money to B's money is 5:7. A gives ₹20 to B, and the ratio becomes 3:5. How much money did A have originally?

Solution: Let amounts be 5x and 7x. $(5x-20)/(7x+20)=3/5$. Thus $25x-100=21x+60$, $x=40$. A had ₹200.

Answer: 0

Vault Code (Round 5): 3870

Round 6

Section 1 — Linear Equations

Solve for x: $11 - 2(x - 1) = 3x - 2$

Answer: 3

Section 2 — Percentages

80% of a number is 64. Find 5% of that number.

Solution: Number = 80, 5% = 4

Answer: 4

Section 3 — Grid Direction & Distance

Starting at her house facing North, Priya walks a sequence of legs, turning left/right relative to her current facing each time: Walks 9 km (facing North). Turns left, walks 7 km. Turns right, walks 3 km. Turns left, walks 9 km.

Turns left, walks 1 km. What is the shortest distance (km, rounded to nearest whole number) between her final position and the start? (If more than 9, give the last digit.)

Solution: Tracking heading and (x,y): final position = (-16,11). Distance = $\sqrt{(-16)^2 + 11^2} = 19.42 \rightarrow$ rounds to 19 $\rightarrow$ last digit 9.

Answer: 9

Section 4 — Ratio & Mixture

Rice and dal are mixed in the ratio 7:3. If 8 kg of the mixture is removed and replaced by 8 kg of dal, the ratio becomes 7:5. Find the original quantity of the mixture.

Solution: In 8 kg removed, rice=5.6 kg and dal=2.4 kg. After replacing it with 8 kg of dal:

$$\text{Rice} = 0.7T - 5.6$$

$$\text{Dal} = 0.3T - 2.4 + 8 = 0.3T + 5.6$$

Set the new ratio to 7:5:

$$(0.7T - 5.6)/(0.3T + 5.6) = 7/5$$

$$5(0.7T - 5.6) = 7(0.3T + 5.6)$$

$$3.5T - 28 = 2.1T + 39.2$$

$$1.4T = 67.2$$

$$T = 48 \text{ kg}$$

Answer: 8

Vault Code (Round 6): 3498

Round 7

Section 1 — Linear Equations

Solve for x: $\frac{x}{2} + \frac{x}{3} = 5$

Answer: 6

Section 2 — Percentages

A number increased by 25% gives 40. Find 10% of the original number (rounded).

Solution: Original = 32, 10% = $3.2 \approx 3$

Answer: 3

Section 3 — Grid Direction & Distance

Starting at her house facing North, Priya walks a sequence of legs, turning left/right relative to her current facing each time: Walks 1 km (facing North). Turns left, walks 5 km. Turns right, walks 8 km. Turns left, walks 1 km.

Turns left, walks 3 km. Turns right, walks 6 km. Turns left, walks 2 km. What is the shortest distance (km, rounded to nearest whole number) between her final position and the start? (If more than 9, give the last digit.)

Solution: Tracking heading and (x,y): final position = (-12,4). Distance = $\sqrt{-12^2 + 4^2} = 12.65 \rightarrow$ rounds to 13

$\rightarrow$ last digit 3.

Answer: 3

Section 4 — Ratio & Mixture

A class has boys and girls in the ratio 3:2. If 10 boys leave and 10 girls join, the ratio becomes 1:1. Find the original number of students.

Solution: Let boys=3x, girls=2x. $3x-10=2x+10$, so $x=20$. Total=5x=100.

Answer: 0

Vault Code (Round 7): 6330

Round 8

Section 1 — Linear Equations

Solve for x: $5(x - 2) + 3 = 3x + 1$

Answer: 4

Section 2 — Percentages

A number decreased by 20% becomes 32. Find 10% of the original number.

Solution: Original = 40, 10% = 4

Answer: 4

Section 3 — Grid Direction & Distance

Starting at her house facing North, Priya walks a sequence of legs, turning left/right relative to her current facing each time: Walks 1 km (facing North). Turns left, walks 8 km. Turns right, walks 7 km. Turns left, walks 5 km. Turns right, walks 4 km. Turns right, walks 1 km. What is the shortest distance (km, rounded to nearest whole number) between her final position and the start? (If more than 9, give the last digit.)

Solution: Tracking heading and (x,y): final position = (-12,12). Distance = $\sqrt{(-12)^2 + 12^2} = 16.97 \rightarrow$ rounds to 17 $\rightarrow$ last digit 7.

Answer: 7

Section 4 — Ratio & Mixture

A juice mixture contains syrup and water in the ratio 2:5. If 6 L of water is removed, the ratio becomes 1:2. Find the original quantity of the mixture.

Solution: Let syrup=2x, water=5x. $2x/(5x-6)=1/2$, so $4x=5x-6$, $x=6$. Total=7x=42 L.

Answer: 2

Vault Code (Round 8): 4472

Round 9

Section 1 — Linear Equations

Solve for x: $(x + 3)/2 - (x - 1)/3 = 3$

Answer: 7

Section 2 — Percentages

A student scores 72% in an exam out of 50 marks. Find 10% of the marks obtained (rounded).

Solution: Score = 36, 10% = $3.6 \approx 4$

Answer: 4

Section 3 — Grid Direction & Distance

Starting at her house facing North, Priya walks a sequence of legs, turning left/right relative to her current facing each time: Walks 9 km (facing North). Turns right, walks 8 km. Turns left, walks 3 km. Turns right, walks 3 km. Turns left, walks 8 km. Turns right, walks 3 km. What is the shortest distance (km, rounded to nearest whole number) between her final position and the start? (If more than 9, give the last digit.)

Solution: Tracking heading and (x,y): final position = (14,20). Distance = $\sqrt{14^2 + 20^2} = 24.41 \rightarrow$ rounds to 24 $\rightarrow$ last digit 4.

Answer: 4

Section 4 — Ratio & Mixture

Two numbers are in the ratio 5:8. If 15 is added to each number, their ratio becomes 2:3. Find the smaller number.

Solution: Let numbers = $5x, 8x$. $(5x+15)/(8x+15) = 2/3$. $15x+45 = 16x+30$, $x=15$. Smaller = 75.

Answer: 5

Vault Code (Round 9): 7445

Round 10

Section 1 — Linear Equations

Solve for x: $2x - 3(x - 2) = -1$

Answer: 7

Section 2 — Percentages

A number is increased by 10% twice, becoming 96.8. Find 5% of the original number (rounded).

Solution: Original = 80, 5% = 4

Answer: 4

Section 3 — Grid Direction & Distance

Starting at her house facing North, Priya walks a sequence of legs, turning left/right relative to her current facing each time: Walks 4 km (facing North). Turns left, walks 1 km. Turns right, walks 6 km. Turns right, walks 4 km. Turns left, walks 5 km. What is the shortest distance (km, rounded to nearest whole number) between her final position and the start? (If more than 9, give the last digit.)

Solution: Tracking heading and (x,y): final position = (3,15). Distance = $\sqrt{3^2 + 15^2} = 15.30 \rightarrow$ rounds to 15
 $\rightarrow$ last digit 5.

Answer: 5

Section 4 — Ratio & Mixture

A basket contains apples and oranges in the ratio 4:5. After 8 apples are removed and 5 oranges are added, the ratio becomes 2:3. Find the original number of fruits.

Solution: Let apples=4x, oranges=5x. $(4x-8)/(5x+5)=2/3$. $12x-24=10x+10$, $x=17$. Original total=9x=153.

Answer: 3

Vault Code (Round 10): 7453

Round 11

Section 1 — Linear Equations

Solve for x: $6(x - 1) - 4(x - 2) = x + 3$

Answer: 1

Section 2 — Percentages

Cost price of an item is ₹20. It is marked up 50% and then sold at a 40% discount. Find 10% of the final selling price (rounded).

Solution: Final price = 18, 10% = $1.8 \approx 2$

Answer: 2

Section 3 — Grid Direction & Distance

Starting at her house facing North, Priya walks a sequence of legs, turning left/right relative to her current facing each time: Walks 1 km (facing North). Turns left, walks 2 km. Turns left, walks 9 km. Turns right, walks 4 km. Turns right, walks 5 km. Turns right, walks 8 km. What is the shortest distance (km, rounded to nearest whole number) between her final position and the start? (If more than 9, give the last digit.)

Solution: Tracking heading and (x,y): final position = (2,-3). Distance = $\sqrt{2^2 + 3^2} = 3.61 \rightarrow$ rounds to 4.

Answer: 4

Section 4 — Ratio & Mixture

A group of red and green balls is in the ratio 3:4. If 6 green balls are removed, the ratio becomes 3:2. How many balls were originally there?

Solution: Let red=3x, green=4x. $3x/(4x-6)=3/2$, so $6x=12x-18$, $x=3$. Total=7x=21.

Answer: 1

Vault Code (Round 11): 1241

Round 12

Section 1 — Linear Equations

Solve for x: $3(x + 2) = 2(x + 4) + 1$

Answer: 3

Section 2 — Percentages

45% of x is 27. Find 10% of x, then subtract 1.

Solution: $x = 60$, $10\% = 6$, $6 - 1 = 5$

Answer: 5

Section 3 — Grid Direction & Distance

Starting at her house facing North, Priya walks a sequence of legs, turning left/right relative to her current facing each time: Walks 8 km (facing North). Turns right, walks 3 km. Turns left, walks 8 km. Turns left, walks 3 km.

Turns right, walks 7 km. What is the shortest distance (km, rounded to nearest whole number) between her final position and the start? (If more than 9, give the last digit.)

Solution: Tracking heading and (x,y): final position = (0,23). Distance = $\sqrt{0^2 + 23^2} = 23.00$ -> rounds to 23

-> last digit 3.

Answer: 3

Section 4 — Ratio & Mixture

A prize is divided among A, B and C in the ratio 2:3:5. If C receives ■240 more than A, what is the total prize?

Solution: Difference $C - A = 3$ parts = ■240, so 1 part = ■80. Total parts = 10, total prize = ■800.

Answer: 0

Vault Code (Round 12): 3530

Round 13

Section 1 — Linear Equations

Solve for x: $3x - 2(x - 3) = 3x - 2$

Answer: 4

Section 2 — Percentages

75% of a number equals 9. Find 10% of that number (rounded).

Solution: Number = 12, 10% = $1.2 \approx 1$

Answer: 1

Section 3 — Grid Direction & Distance

Starting at her house facing North, Priya walks a sequence of legs, turning left/right relative to her current facing each time: Walks 1 km (facing North). Turns right, walks 7 km. Turns right, walks 7 km. Turns left, walks 4 km. Turns right, walks 6 km. What is the shortest distance (km, rounded to nearest whole number) between her final position and the start? (If more than 9, give the last digit.)

Solution: Tracking heading and (x,y): final position = (11,-12). Distance = $\sqrt{11^2 + 12^2} = 16.28 \rightarrow$ rounds to 16 $\rightarrow$ last digit 6.

Answer: 6

Section 4 — Ratio & Mixture

A mixture contains chemical A and B in the ratio 5:2. If 9 L of B is added, the ratio becomes 5:3. Find the original quantity of A.

Solution: Let $A=5x$, $B=2x$. $5x/(2x+9)=5/3$, so $15x=10x+45$, $x=9$. $A=45$ L.

Answer: 5

Vault Code (Round 13): 4165

Round 14

Section 1 — Linear Equations

Solve for x: $3(2x + 1) - 2(3x - 1) = x - 1$

Answer: 6

Section 2 — Percentages

A price rises 10%, then falls 10%. The final price is 99. Find 1% of the original price.

Solution: Original = 100, 1% = 1

Answer: 1

Section 3 — Grid Direction & Distance

Starting at her house facing North, Priya walks a sequence of legs, turning left/right relative to her current facing each time: Walks 5 km (facing North). Turns right, walks 1 km. Turns right, walks 2 km. Turns left, walks 4 km. Turns left, walks 1 km. What is the shortest distance (km, rounded to nearest whole number) between her final position and the start? (If more than 9, give the last digit.)

Solution: Tracking heading and (x,y): final position = (5,4). Distance = $\sqrt{5^2 + 4^2} = 6.40 \rightarrow$ rounds to 6.

Answer: 6

Section 4 — Ratio & Mixture

The ratio of boys to girls in a club is 5:4. If 10 boys and 2 girls join, the ratio becomes 3:2. Find the original total number of members.

Solution: Let boys=5x, girls=4x. $(5x+10)/(4x+2)=3/2$. $10x+20=12x+6$, $x=7$. Original total=9x=63.

Answer: 3

Vault Code (Round 14): 6163

Round 15

Section 1 — Linear Equations

Solve for x: $(3x - 2)/4 = (x + 2)/2 - 1$

Answer: 2

Section 2 — Percentages

30% of a number is 24. Find 10% of that number.

Solution: Number = 80, 10% = 8

Answer: 8

Section 3 — Grid Direction & Distance

Starting at her house facing North, Priya walks a sequence of legs, turning left/right relative to her current facing each time: Walks 2 km (facing North). Turns left, walks 3 km. Turns left, walks 8 km. Turns right, walks 1 km. Turns right, walks 5 km. Turns right, walks 2 km. What is the shortest distance (km, rounded to nearest whole number) between her final position and the start? (If more than 9, give the last digit.)

Solution: Tracking heading and (x,y): final position = (-2,-1). Distance = $\sqrt{(-2)^2 + (-1)^2} = 2.24 \rightarrow$ rounds to 2.

Answer: 2

Section 4 — Ratio & Mixture

A shop has pens and pencils in the ratio 3:5. After selling 12 pencils and adding 6 pens, the ratio becomes 3:4. Find the original number of items.

Solution: Let pens=3x, pencils=5x. $(3x+6)/(5x-12)=3/4$. $12x+24=15x-36$, $x=20$. Total=8x=160.

Answer: 0

Vault Code (Round 15): 2820